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SUBJECT CODE NO: RNP- 8006
FACULTY OF SCIENCE AND TECHNOLOGY
F.E. (Group B) CSE / AI & ML / EEE (NEP) Sem - I
Examination June/July 2025
Principles of Programming Language

[Time: 3 Hours]**[Max. Marks: 60]**

Please check whether you have got the right question paper.

N. B

- 1) Q. no. 1 is compulsory.
- 2) Solve any four questions from Q.no. 2, 3, 4, 5, 6 and 7.

SECTION A**Q1 Solve any ten.****20**

A. Semantic of the program means

- a) Format of a program
- b) meaning of the program
- c) simply content of the program
- d) none of these

B. In C, reference is declare using symbol

- a) *
- b) &
- c) &&
- d) !

C. What is output of the following code

```
int main(void)
{
    char name [20] ="ppl";
    name [] =10;
    return 0;
}
```

- a) Name will contain value 10.
- b) P will contain value 10
- c) Syntax Error
- d) None of these

D. Which is longest scope in the following code

```
#include<stdio.h>
int x;
int main ()
{
    int y;
    fun ();
    return 0;
}
void fun ()
{
    int z;
}
```

- a) x
- b) y
- c) both a and b.
- d) z

E. What will be the output of the following C code?

```
void main ()
{
    int a=10, b=20;
    char x='l', y='0';
    if(a,b,x,y)
    {
        printf("EXAM");
    }
}
```

- a) XAM is printed b) exam is printed
c) Compiler Error d) Nothing is printed

F. What will be the output of the following arithmetic expression?

$5+3*2\%10-8*6$

- a) -37 b)-42 c)-32 d) -28

G. What will be the output of the following statements?

int a = 4, b = 7, c; c = a = b; printf ("%i", c);

- a) 0 b) error c) 1 d) garbage value

H. What will be the output of the following C code?

```
#include <stdio.h>
void main ()
{
    int x = 0;
    int *ptr = &x;
    printf ("%d\n", *ptr);
}
```

- a) Address of x b) Junk value
c) 0 d) Run time error

I. In C, Functions are primarily used for

- a) Decision making b) Variable declaration
c) Code organization and reusability d) printing output

J. What will be the output of the following C code?

```
#include <stdio.h>
int main ()
while ()
printf("In while loop ");
printf("After loop\n");
}
```

- a) In while loop after loop b) After loop
c) Compile time error d) Infinite loop

K. Union differs from structure in the following way

- a) All members are used at a time
b) Only one member can be used at a time
c) Union cannot have more members
d) Union initialized all members as structure

- L. What are the different ways to initialize an array with all elements as zero?
- `int array [5] = {};`
 - `int array [5] = {0};`
 - `int a = 0, b=0, c = 0;`
`int array [5] = {a, b, c};`
 - All of the mentioned
- M. Which of the following is the correct definition of Computer?
- Computer is a machine or device that can be programmed to perform arithmetical or logic operation sequences automatically
 - Computer understands only binary language which is written in the form of 0s & 1s
 - Computer is a programmable electronic device that stores, retrieves, and processes the data
 - All of the mentioned

SECTION B

- | | | |
|-----------|--|----------|
| Q2 | A) List and explain Input and Output devices of computer | 5 |
| | B) Explain any two program translators with the help of neat diagram | 5 |
| Q3 | A) Explain types constants in C. | 5 |
| | B) Explain the steps of program development | 5 |
| Q4 | A) Write a program to implement simple calculator using switch-case in C. | 6 |
| | B) Write a program to read a number from user and display sum of its digits. | 4 |
| Q5 | A) Write a program in C to create a structure having named as Bank_Acc consists Customer name, Account number, address, and Balance as its members. Use array of structure to read the details for five bank customer from user and then display the data entered by the user on Screen. | 5 |
| | B) Define union in C. Also state the difference between union and structure. | 5 |
| Q6 | A) Write a program to accept marks of 5 subject of student from user calculate and display total and average marks also display grade according to the average marks (A+ grade for avg marks between 91 to 100, A grade for avg marks between 81 to 90, B+ grade for avg marks between 71 to 80, B grade for avg marks between 61 to 70, C grade for avg marks between 41 to 60) | 5 |
| | B) Explain steps to implement user-defined functions in C program. | 5 |
| Q7 | A) Explain Multidimensional array in detail | 5 |
| | B) Explain any one user defined data type in C | 5 |